OBONE ESSONO Hegirziana Heva

Master's Degree in Medical Physics

Seeking a 5 to 6-month Internship

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LinkedIn | **GitHub Website**

Education & Qualifications

- **M.Sc. in Health Engineering - Specialization in Medical Physics** — Toulouse, France
  University of Toulouse — *2026 - Present*
  - Dosimetry, Medical Imaging Techniques **(MRI, PET, SPECT, Ultrasound)**, Radiotherapy, Radiation Protection, Radiation-Matter Interactions, Monte Carlo Simulation, Image Processing, Programming
- **Bachelor's Degree in Electronics, Energy, Automatics (EEA) - Health Track** — Toulouse, France
  University of Toulouse — *2025*

Work Experience

- **Medical Physics Intern (2 months)** — Libreville, Gabon
  Akanda Cancer Institute (ICA) — *Jun 2026 - Jul 2026*
  - **Focus** : Clinical immersion in **Radiotherapy** and **Nuclear Medicine** departments.
  - Daily, weekly, and monthly quality assurance (QA) testing on radiotherapy equipment (**LINAC, CT simulator, ISOMED dose calibrator**).
  - Hands-on practice with **Treatment Planning Systems (TPS)**; Organs at Risk (OAR) delineation, target volume contouring (CTV, PTV), field design, particle and energy selection.
- **Research Intern (1 month)** — Toulouse, France
  LAPLACE Laboratory — PRHE Team — *Jun 2025 - Jul 2025*
  - **Focus & Mission** : Experimental study of cold plasma / microorganism interactions - Evaluation of the efficiency of **reactive species (ROS/RNS)** generated by plasma discharge for biological decontamination (sterilization).

Academic Projects

- **Master 2 - Radiobiology, Dosimetry, Monte Carlo Simulation** — Toulouse, France
  University of Toulouse - Inserm — *Ongoing*
- **Master 1 - Brain Metabolic Quantification - MATLAB & SPM12** — Toulouse, France
  Toulouse NeuroImaging Center (ToNIC) - Inserm — *Mar 2026 - May 2026*
  - **Focus** : **PET Quantification** and analysis of regional metabolic deficits in Alzheimer's disease.
  - Preprocessing of 3D brain volumes using SPM (spatial coregistration, normalization, and resampling via trilinear interpolation), **SUV and SUVR calculation** using the cerebellum as the reference region.
  - Quantitative metabolic analysis highlighting key disease regions (temporal, parietal, and frontal lobes) with profile comparison (healthy subjects vs. cognitive impairment / Alzheimer's).
- **Master 1 - Medical Image Interface in C++** — Toulouse, France
  University of Toulouse — *Feb 2026 - May 2026*
  - **Focus** : Design of an interactive interface for 3D visualization and processing of medical images (**DICOM**).
  - 2D Gaussian filtering, Look-Up Tables (LUT) for contrast adjustment and color mapping.
  - Simulation of a **radiotherapy beam** and reconstruction of **Digitally Reconstructed Radiographs (DRR)**.
  - Segmentation of anatomical structures (bones/tissues) and smooth rendering integrated into the UI.
- **Bachelor Year 3 - Vectorization & Huntington's Disease** — Toulouse, France
  SOFTMAT Laboratory — *Sep 2024 - Jan 2025*
  - **Study Focus :** **Vectorization techniques** and targeted therapy in the treatment of neurodegenerative diseases.

Technical & Computer Skills

- **Neuroimaging & Imaging :** **SPM12**, MRIcron/MRIcroGL, PET/MRI processing (DICOM), SUV/SUVR calculation, spatial normalization (MNI space).
- **Programming : Python**, **C, C++, MATLAB**.
- **Environments :** VS Code, PyCharm, Qt Creator, Linux.

Soft Skills, Languages & Honors

- **Soft Skills :** Scientific rigor, analytical & synthesis skills, teamwork, autonomy.
- **Honor :** Merit-based High School Baccalaureate Scholarship.
- **French :** Native speaker | **Spanish :** B1 (Intermediate)
- **English :** C2 (EF SET Certificate) | **Mandarin :** A2

References

- **LASSY TATY Aurélia**
  Medical Physicist - ICA
  aurelialt89@gmail.com
- **Dr. MABIKA NDJEMBIDOUMA Beau Conrad**
  Head of Medical Physics Department - ICA
  mabikabeaud@yahoo.fr

Interests

- **Board games, Sudoku, Puzzles**
- **Science Fiction & Crime Novels**
- **Foreign cultures and languages**
- **Scientific innovations and advancements**